

3.1 An Introduction to AS Level Organic Chemistry

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.1 An Introduction to AS Level Organic Chemistry
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which pairs of homologous series do not have the same C:H ratio in their general formulae?

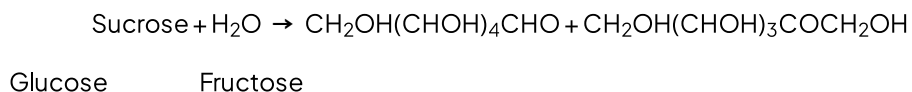
- A. Aldehydes and ketones
- B. Alkanes and alkenes
- C. Carboxylic acids and esters
- D. Alkenes and aldehydes

[1 mark]

Question 2

The disaccharide sucrose, $C_{12}H_{22}O_{11}$, is found in biscuits and cakes.

Inside the body, sucrose is hydrolysed to produce glucose and fructose:



Which statements about the hydrolysis products above are correct?

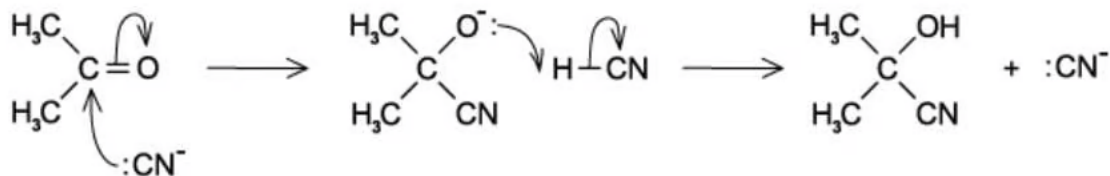
- 1 Glucose and fructose have structural isomers.
- 2 Glucose and fructose are ketones.
- 3 Glucose is a carboxylic acid.
- 4 Glucose and fructose have optical isomers.

- A. 1 and 2
- B. 2 and 4
- C. 1 and 4
- D. 1, 2 and 4

[1 mark]

Question 3

The following mechanism shows hydrogen cyanide reacting with propanone.



Which of the following statements is incorrect?

- A. Heterolytic bond breaking is involved.
- B. CN^- is an electrophile.
- C. This is an addition reaction.
- D. Propanone is a ketone.

[1 mark]

Question 4

A molecule can have optical isomerism and geometrical isomerism.

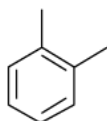
Which of the compounds below show both?

- A. $\text{CH}_3\text{CH}_2\text{CHBrCH}=\text{CHBr}$
- B. $\text{CH}_3\text{CBr}=\text{CBrCH}_3$
- C. $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_3$
- D. $\text{CH}_3\text{CHBrCH}=\text{CH}_2$

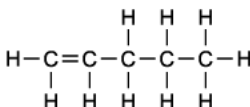
[1 mark]

Question 5

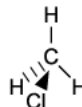
Which of the following molecules are hydrocarbons?



1



2



3



4

- A. All of the above
- B. 1 and 2
- C. 1, 2 and 3
- D. 4 only

[1 mark]

Question 6

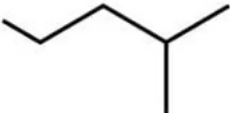
Which of the following statements correctly describes 'electrophilic addition'?

- A. The organic molecule acts as a nucleophile and becomes more saturated.
- B. The organic molecule acts as an electrophile and becomes more saturated.
- C. The organic molecule acts as a nucleophile and becomes less saturated.
- D. The organic molecule acts as an electrophile and becomes less saturated.

[1 mark]

Question 7

What is the correct formula for 1-fluoro-3-methyl-2-butene?

A	Structural	$\text{CH}_2\text{FCH}_2\text{C}(\text{CH}_3)\text{CH}_3$
B	Empirical	$\text{C}_5\text{H}_9\text{F}$
C	Displayed	$ \begin{array}{ccccccc} & & & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - & \text{C} = & \text{C} - & \text{C} - & \text{C} - & \text{C} - \text{F} \\ & & & & & & \\ & & \text{H} & \text{H} & \text{H} & \text{H} & \end{array} $
D	Skeletal	

[1 mark]

Question 8

Which of these organic compounds would undergo free radical substitution?

- 1 Ethane
- 2 Fluoroethane
- 3 Ethene
- 4 Ethanal

A. 1 only

B. 1 and 2

C. 1, 2 and 3

D. All of the above

[1 mark]

Question 9

Which of these compounds would act as a nucleophile?

- A. C_2H_6
- B. H^+
- C. OH^-
- D. $\text{Cl}^\bullet$

[1 mark]

Question 10

A student is building a non-cyclic, chiral alkane from a molymod kit. What is the lowest number of carbon atoms they will need to include in the model?

- A. 6
- B. 11
- C. 9
- D. 7

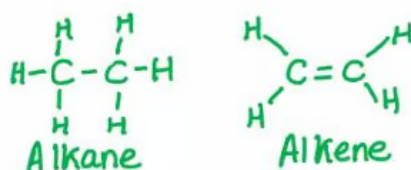
[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- A homologous series is a group with the same functional group and similar chemical properties.
- Alkanes and alkenes are different homologous series as they have a different functional group.
- Alkanes have single carbon-carbon bonds and the general formula C_nH_{2n+2}
- Alkenes have a double carbon-carbon bond and the general formula C_nH_{2n}



A is incorrect as aldehydes and ketones both have the general formula of $C_nH_{2n}O$.

C is incorrect as the general formula for carboxylic acids is $C_nH_{2n+1}COOH$. The general formula for an ester is $R'COOR$. An ester is a carboxylic acid with the hydroxy group of the carboxyl group replaced with a hydrocarbon chain.

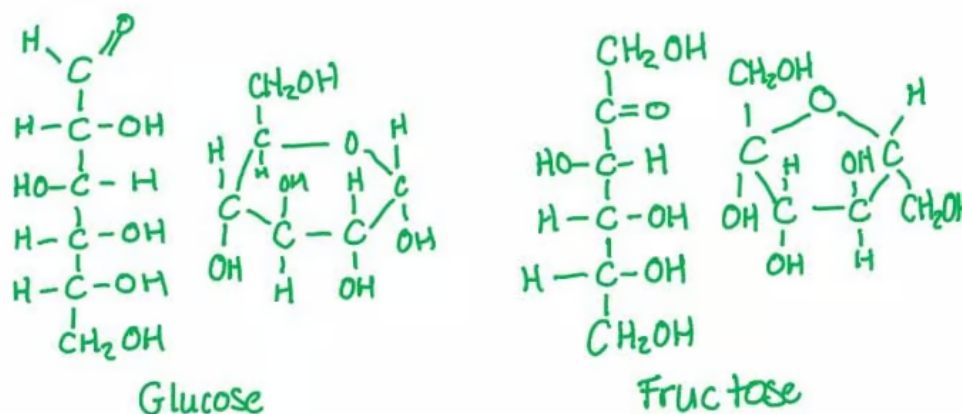
D is incorrect as alkenes and aldehydes have the same carbon-hydrogen ratio in their general formula. Alkenes have the general formula C_nH_{2n} and aldehydes $C_nH_{2n}O$.

2

The correct answer is **C** because:

- Structural isomers are molecules with the same formula but with the atoms in a different order.

- Both glucose and fructose have a chain and a ring structure.



Optical isomers are isomers that have the same atoms and bonds but different spatial arrangements of the atoms; the isomers are mirror images of each other and cannot be superimposed over one another (they are enantiomers). Fructose and glucose have 16 optical isomers each, as each molecule has 4 chiral carbons.

3

The correct answer is **B** because:

- The reaction between hydrogen cyanide and propanone is a nucleophilic addition reaction.
- A nucleophile is an electron-pair donor.
- An electrophile is an electron pair acceptor.
- In this reaction, CN^- is donating a pair of electrons (as shown by the arrow) and the carbon in propanone is accepting them.

A is incorrect as this is heterolytic bond breaking. Only one atom is getting both the electrons from the lone pair, making this heterolytic.

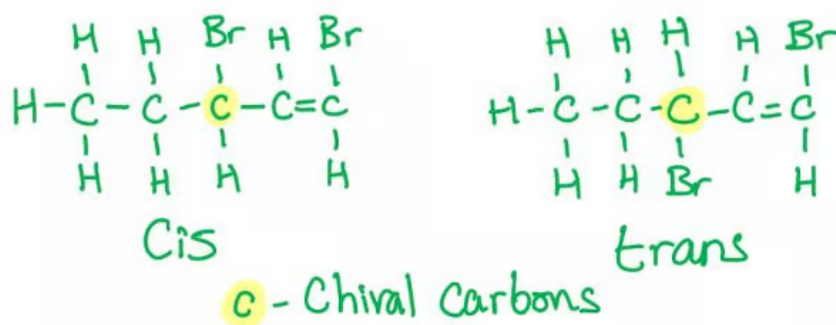
C is incorrect as this is an example of a nucleophilic addition reaction.

D is incorrect as propanone is an example of a ketone.

4

The correct answer is **A** because:

- Optical isomers are isomers that have the same atoms and bonds but different spatial arrangements of the atoms; the isomers are mirror images of each other and cannot be superimposed over one another (they are enantiomers).
 - These are formed when a molecule has a chiral centre.
 - A chiral centre is identified by a carbon atom with 4 different groups/atoms bonded to it.
- Two types of geometric isomers are possible due to the restriction surrounding the C = C double bond.
 - Cis isomers have the functional group on the same side of the carbon chain.
 - Trans isomers have the functional groups on opposite sides of the carbon chain.



B is incorrect as this has cis-trans but no chiral carbons, therefore no optical isomerism.

C is incorrect as this has cis-trans but no chiral carbons and therefore no optical isomerism.

D is incorrect as this doesn't have cis-trans isomerism.

5

The correct answer is **B** because:

- A hydrocarbon is a molecule with only carbon and hydrogen.
- Molecule 1 shows a skeletal structure which only contains hydrogen and carbon; we know this as there are no other elements shown in the molecule.

- Molecule 2 shows the displayed formula of a hydrocarbon.

Molecule 3 and 4 both have elements that are not either hydrogen or carbon.

6

The correct answer is **A** because:

- If a reaction is electrophilic, this means that the inorganic reagent is acting as an **electrophile** (electron pair accepting)

Cis Trans
c - Chiral carbons

B is incorrect as this has cis-trans but no chiral carbons, therefore no optical isomerism.

C is incorrect as this has cis-trans but no chiral carbons and therefore no optical isomerism.

D is incorrect as this doesn't have cis-trans isomerism.

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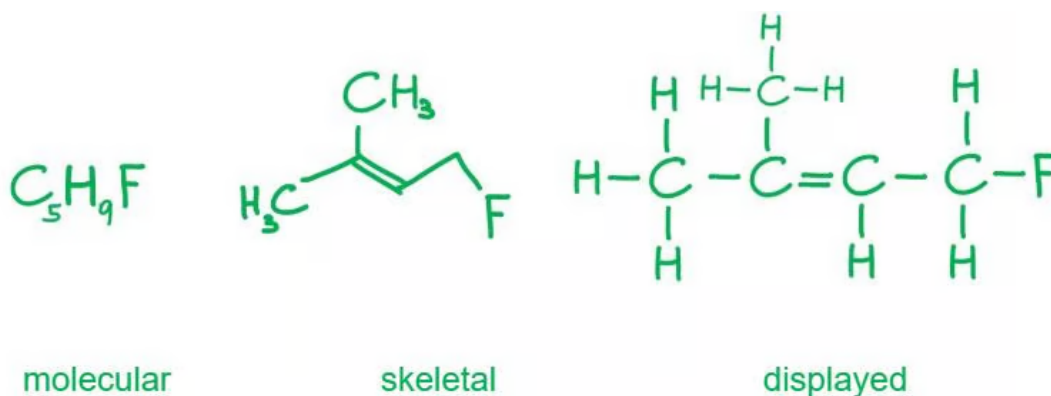
The correct answer is **A** because:

- If a reaction is electrophilic, this means that the inorganic reagent is acting as an **electrophile** (electron pair accepting).
- Therefore, the organic reactant is acting as a **nucleophile** (electron pair donating) – this is usually because the organic molecule is an alkene and one of the electron pairs in the double bond is donated to the inorganic reagent, eg. HBr
- In an addition reaction, the organic molecule **gains atoms**.
- In the case of an electrophilic addition reaction and **alkene** is turned into a halogeno**alkane** and therefore becomes **more** saturated.

7

The correct answer is **B** because:

- When naming a chemical using the IUPAC rules, the following steps are used:
 - Find the longest chain of carbons; this is the parent chain.
 - Identify all the groups (substituents) coming off the parent chain.
 - Number the carbons from the end that give the substituents the lowest numbers.
 - If there are two or more substituents, they are listed in alphabetical order using the base name (ignore prefixes such a di or tri).
- The structures of 1-fluoro-3-methyl-2-butene are:



A is incorrect as this is not a correct formula as there are 10 H shown in the structure.

C is incorrect as the name shows that there is a methyl group on carbon 3 so this displayed formula is wrong.

D is incorrect as this skeletal formula is not showing the fluorine or the double bond, and the branch is from the wrong carbon.

8

The correct answer is **B** because:

- Using UV light, free radical substitution is breaking a carbon-hydrogen bond in alkanes.

- Free radicals are atoms or groups of atoms which have a single unpaired electron – radicals are not charged despite having an unpaired electron.
- Free radicals are formed from homolytic fission (where each atom gets one of the two shared electrons to form a covalent bond).

Option 3 and 4 both contain double bonds so these would undergo free radical addition.

9

The correct answer is **C** because:

- A nucleophile is an electron pair donor; they are either negative ions or have a lone pair of electrons capable of being donated.
- The nucleophile in the list above would be OH^- .

A is incorrect as ethane has no lone pairs so will not act as a nucleophile.

B is incorrect as a hydrogen ion would be an electrophile, not a nucleophile. An electrophile is an electron pair acceptor. They are ions with a full or partial positive charge on the atom or molecule.

D is incorrect as metal atoms neither act as nucleophiles or electrophiles.

10

The correct answer is **D** because:

- For the carbon to be chiral, it must have 4 different substituents (groups) attached.
- The question is asking about an alkane so the substituents; hydrogen, methyl, ethyl, propyl or isopropyl are the smallest.
- The smallest number of carbons would be 7.

